

Data Converters

Exercise 07 – Pipeline ADC

Task I – Analysis of a Pipeline ADC

The block diagram of a 3-stage pipeline ADC is shown in Figure 1.

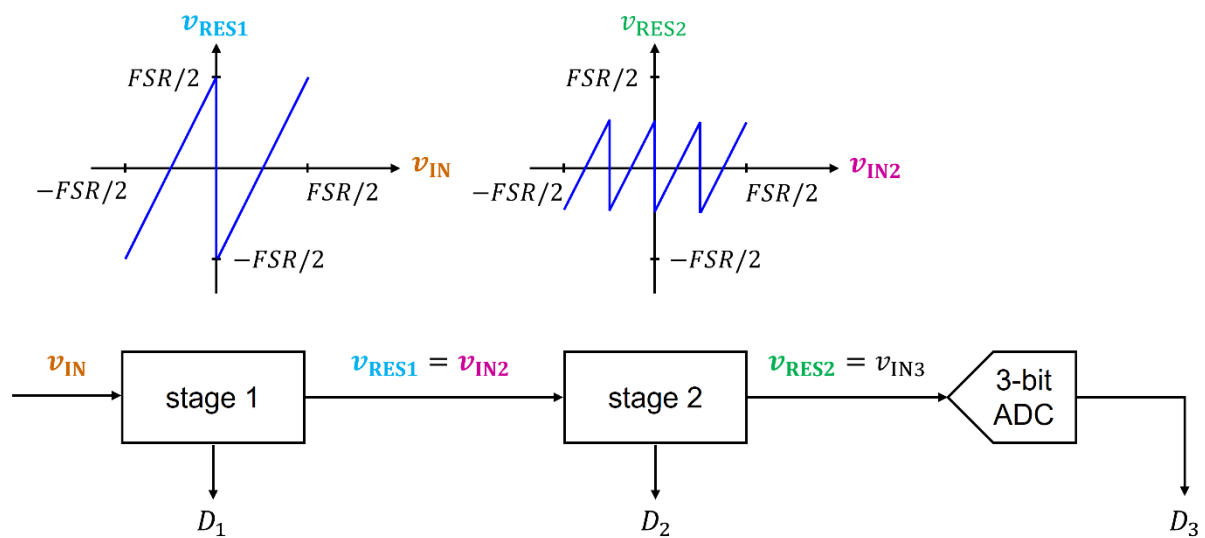


Figure 1: Pipeline ADC.

- a. Determine the resolution of the sub-ADC of the first and the second stage.

stage 1: 1-bit ADC

stage 2: 2-bit ADC

- b. Determine $v_{RES1}(v_{IN}, e_{q1})$ $v_{RES2}(v_{IN2}, e_{q2})$, where q_1 and q_2 are the quantization error of the sub-ADC of the first and the second stage.

$$v_{RES1} = -G_1 e_{q1}$$

$$v_{RES2} = -G_2 e_{q2}$$

- c. Determine the gain G_1 and G_2 of the first and the second stage.

$$G_1 = G_2 = 2$$

- d. Determine the resolution of the pipeline ADC.

5-bit pipeline ADC

- e. Compare the resolution of the pipeline ADC with the sum of the resolutions of its sub-stages. What is the explanation for the inconsistency?

5-bit pipeline ADC vs. 6-bit sum of the resolutions

explanation: scaling of stage 2 equals 2 although it could be 4 → 1-bit less resolution

- f. Determine the maximum tolerable error of the sub-ADC of the first and the second stage.

hint: The input signal of the final ADC must be limited to its input full-scale range.

stage 2: $\pm FSR/8$

stage 1: $\pm FSR/16$

- g. Considering the results of task I.f, briefly explain why the ADC is still functional for

$$-FSR/2 - FSR/16 \leq v_{IN} \leq FSR/2 + FSR/16$$

and

$$-FSR/2 - FSR/8 \leq v_{IN2} \leq FSR/2 + FSR/8$$

although stage 1 and stage 2 are overloaded.

All quantization errors except for the quantization error of the final stage are cancelled. → Only the input signal of the final stage must be limited to $\pm FSR/2$ in order not to overload the pipeline ADC.

- h. Consider Figure 1 and derive the transfer characteristic of the ADCs used in stage 1 and stage 2, i.e., D_1 vs. v_{IN} and D_2 vs. v_{IN2} .

stage 1: transition @ zero, output levels $\pm FSR/4$

stage 2: transition @ zero & $\pm FSR/4$, output levels $\pm FSR/8$ and $\pm 3/8 FSR$

- i. Use your results of task I.b, task I.h as well as the transfer characteristic of the 3-bit ADC given in Figure 2 to determine D_1 , D_2 , D_3 and $D_{OUT} = D_1 + D_2/G_1 + D_3/G_1G_2$ in the analog domain for $v_{IN} = 0.126 FSR = 63/500 FSR$. Moreover, determine v_{RES1} and v_{RES2} .

$$D_1 = \frac{1}{4} FSR \quad D_2 = -\frac{1}{8} FSR \quad D_3 = -\frac{3}{16} FSR \quad D_{OUT} = \frac{9}{64} FSR$$

$$v_{RES1} = -\frac{248}{1000} FSR \quad v_{RES2} = -\frac{246}{1000} FSR$$

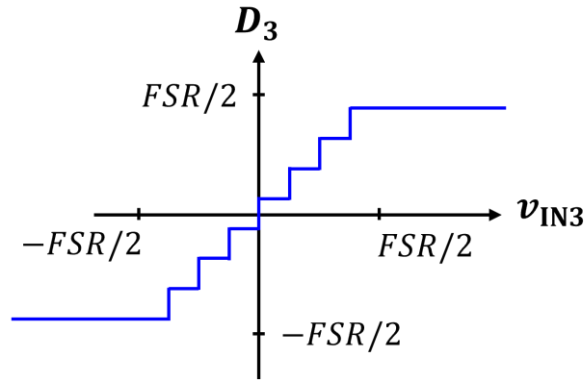


Figure 2: Transfer characteristic of the 3-bit ADC.

- j. Repeat task I.i using the transfer characteristic for the ADC of stage 2 shown in Figure 3 where the transition at $-FSR/4$ shifted to $-FSR/8$.

$$D_1 = \frac{1}{4} FSR \quad D_2 = -\frac{3}{8} FSR \quad D_3 = \frac{5}{16} FSR \quad D_{OUT} = \frac{9}{64} FSR$$

$$v_{RES1} = -\frac{248}{1000} FSR \quad v_{RES2} = \frac{254}{1000} FSR$$

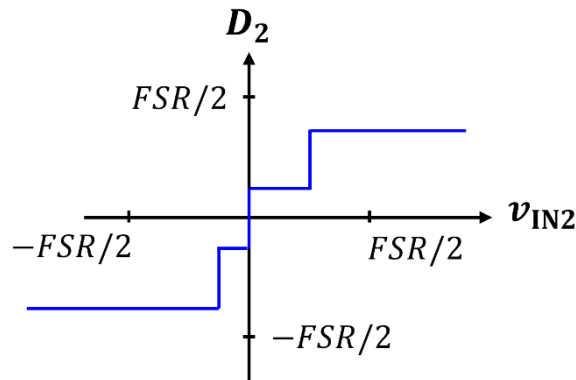


Figure 3: Non-ideal transfer characteristic of the ADC of stage 2.

- k. Repeat task I.i using the transfer characteristic for the DAC of stage 2 shown in Figure 4 where the analog output for code 01 is $-FSR/16$ instead of $-FSR/8$.

$$D_1 = \frac{1}{4} FSR \quad D_2 = -\frac{1}{8} FSR \quad D_3 = -\frac{5}{16} FSR \quad D_{OUT} = \frac{7}{64} FSR$$

$$v_{RES1} = -\frac{248}{1000} FSR \quad v_{RES2} = -\frac{371}{1000} FSR$$

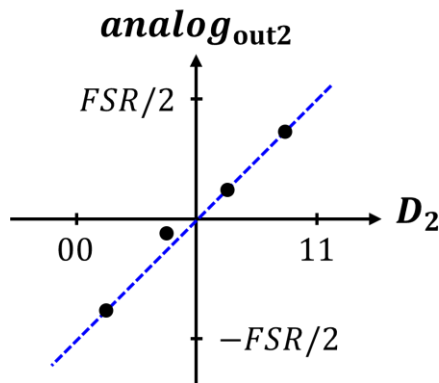


Figure 4: Non-ideal transfer characteristic of the DAC of stage 2.

- I. Comparing your results of task I.j with task I.k: Briefly explain why the pipeline ADC can compensate for the error of the ADC but not of the DAC in stage 2.
- Due to 1-bit redundancy in stage 2, the requirement on the precision of any transition level of the ADC in stage 2 relaxes to $|\pm FSR/8|$.
 - mismatch between ADC and DAC in stage 2 $\rightarrow$ leakage of e_{q2} (2-bit ADC) to the output

Task II – Yet Another Analysis of a Pipeline ADC

The block diagram of an arbitrary pipeline ADC is shown in Figure 5. Unfortunately, the interstage gain between the first stage and the remaining stages summarized in the block backend was implemented as $3/2$ instead of 2 .

The 2 – bit ADC has transition levels at $-FSR/4$, 0 , and $FSR/4$. Moreover, the 2 – bit DAC and backend are ideal and the backend can be assumed to have infinite resolution/precision.

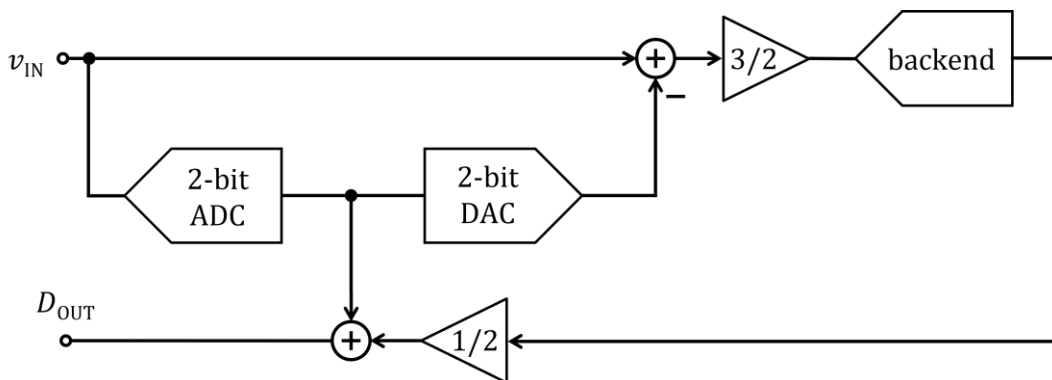
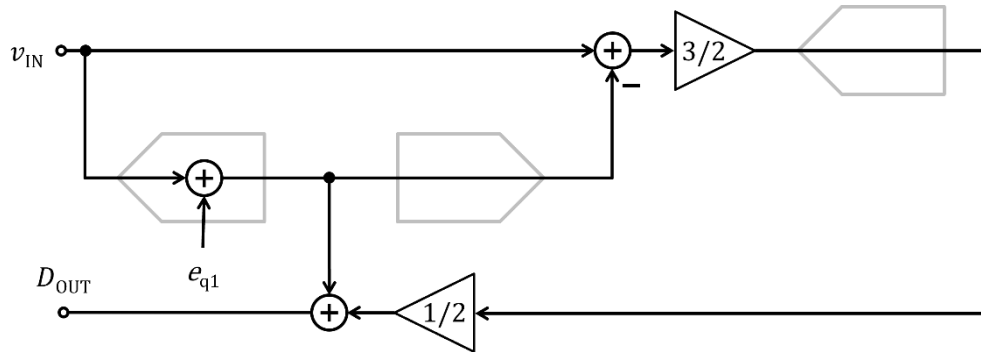


Figure 5: Arbitrary pipeline ADC.

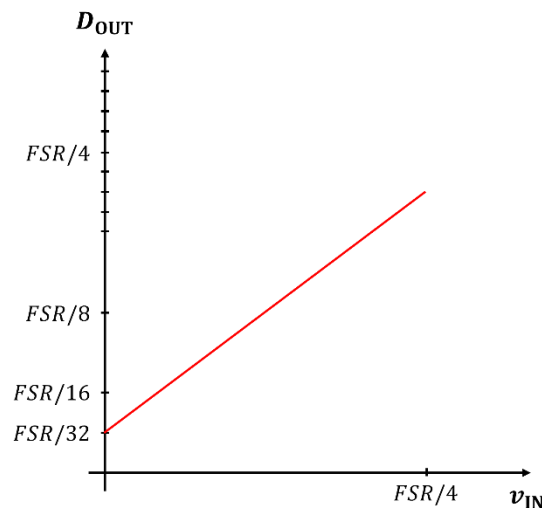
a. Draw the linearized model of the pipeline ADC.



b. Determine D_{OUT} in the analog domain.

$$D_{OUT} = v_{IN} + \frac{1}{4}e_{q1}$$

c. Sketch D_{OUT} versus v_{IN} in the analog domain for one segment of the 2 – bit ADC, e.g., between 0 and $FSR/4$.



d. How large is the slope of the segment?

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{3}{4}$$

e. How large is the jump at the transition points?

$$\Delta = \frac{FSR}{16}$$

Task III – Final Analysis of a Pipeline ADC

A 9-stage pipeline ADC uses a 2-bit flash ADC in the last stage. All other stages are implemented using a circuit block that generates a residue according to Figure 6. All interstage gains are set to x2.

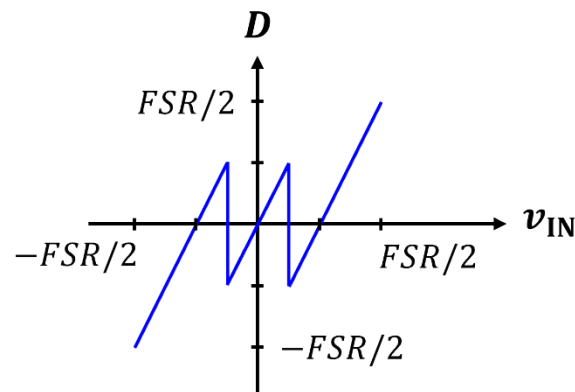


Figure 6: Circuit block used in the pipeline ADC of task III.

- a. Briefly explain why this circuit block is called a 1.5-bit stage.

$$\log_2 3 = \frac{\ln 3}{\ln 2} \approx 1.585$$

- b. Determine the maximum tolerable error of the sub-ADCs.

$$error_{\max} = \pm FSR/8$$

- c. A friend of yours comes up with an idea that he/she considers very clever: In order to save power, all the interstage gains of x2 were removed. Instead, the full-scale ranges were scaled down by a factor x1/2 from stage to stage, i.e., $FSR/2$, $FSR/4$, $FSR/8$, and so on.

Which of the 1.5-bit stages thus becomes most sensitive to sub-ADC errors?

The most sensitive stage is the final 1.5-bit stage (stage 9) since its full-scale reduces to $FSR/2^8$ ($\pm FSR/256$).

Determine the maximum tolerable error of the sub-ADCs.

$$error_{\max} = \pm FSR/8$$